



Effects of Chinese Herbal Medicines on Lipid Metabolism and Immunity Function in Laying Hens

Chunhong Li, Yulong Dong, Ruihua Zhang, Li Wang, Wanyu Shi¹, Tong Xu, Hongliang Zhang²

10.18805/ijar.B-1261

ABSTRACT

Background: Lipid is considered as one of the essential nutrients for both humans and animals. Lipid accumulation within the body may induce several health disorders. Disease is caused easily in aged laying hens. It is necessary to improve lipid metabolism and enhance immunity in laying hens. Traditional Chinese veterinary medicine (TCVM) with no toxicity and low drug residues is becoming more and more popular. The current study aimed to study the effect of Chinese herbal medicines on lipid metabolism and immunity function in chicken to provide basis for clinical reference.

Methods: In this laboratory investigation during 2018-2019, a total of 400 60-week-old hens were randomized into 10 groups to measure lipid metabolism by colorimetry and immunoglobulin by double antibody sandwich ELISA method. A total of 500 28-day-age chickens were randomized into 10 groups to measure immune organ index. The experiment lasted for 42 days.

Result: Our investigations have showed that *Jujube* (ZJM) has significant effects on the count of Serum total cholesterol (TC), Triglyceride (TG), Low-density lipoprotein-c (LDL-C) and High-density lipoprotein-c (HDL-C) ($p < 0.01$, $p < 0.05$), the count of Immunoglobulin-G (IgG) ($p < 0.05$) and the immune organ indexes ($p < 0.05$) in 30 mg/kg group than control group. The present work will be a complementary contribution to the ability of ZJM on improving lipid metabolism and immunity function in chickens.

Key words: *Angelicae sinensis extract*, *Astragalus*, Immunity function, *Jujube*, Lipid metabolism.

INTRODUCTION

Lipid is one of the essential nutrients for both humans and animals, due to its role as an energy source and a provider of essential fatty acids. However, lipid accumulation within the body may induce several health disorders, resulting in a demand to regulate lipid metabolism in laying hens that can be critical for practical production cost and animal welfare. Lipid accumulation is highly linked with hyperlipidemia and characterized by a high level of serum cholesterol (Chen *et al.*, 2008), which results in occurrence and development of cardiovascular diseases (Deogburn *et al.*, 2012; Sun *et al.*, 2015; Sozen and Ozer, 2017). Lipid accumulation in the abdomen is negatively correlated to the reproductive performance and feed efficiency of chickens (Teck *et al.*, 2011; Wen *et al.*, 2018). Lipid accumulation is caused easily in aged laying hens. Reducing blood lipid and sugar levels can effectively delay the development of disease (Manikandan *et al.*, 2017). The immune system is the most prominent line of defense against diseases. The innate immune system is ready to act and can stop infections before they cause disease. Disease is caused easily in aged laying hens. Efforts aiming to maintain birds' health and immune homeostasis comprise the use of dietary supplements (Liu *et al.*, 2014). Therefore, it is necessary to improve lipid metabolism and enhance immunity in laying hens.

The European Union has banned the use of antibiotics as growth-promoting agents in the poultry industry and many countries are restricting the use of antibiotic as growth-promoting agents. Therefore, it is essential to find possible alternatives to antibiotics for growth promotion and

Department of Animal Medicine, Hebei North University, Zhangjiakou 075131, China.

¹College of Veterinary Medicine, Agricultural University of Hebei, Baoding 071001, China.

²College of Veterinary Medicine, Qingdao Agricultural University, Qingdao 266109, China.

Corresponding Author: Tong Xu, Department of Animal Medicine, Hebei North University, Zhangjiakou 075131, China.

Email: xutong1969@126.com

How to cite this article: Li, C., Dong, Y., Zhang, R., Wang, L., Shi, W., Xu, T. and Zhang, H. (2020). Effects of Chinese Herbal Medicines on Lipid Metabolism and Immunity Function in Laying Hens. Indian Journal of Animal Research. 54(10): 1291-1295.

Submitted: 12-03-2020 **Accepted:** 16-07-2020 **Published:** 28-09-2020

improvement in poultry production. Traditional Chinese veterinary medicine (TCVM) with no toxicity and low drug residues is becoming more and more popular.

Jujube (ZJM), with the scientific name of *Ziziphus jujube* Miller, belongs to Rhamnaceae family, mostly as a treatment for tumors and cardiovascular diseases. Antidiabetic and hypolipidaemic effects of *Ziziphus* have been illustrated in streptozocin-induced diabetic rats (Glombitza *et al.*, 1994, Naftali *et al.*, 2009, Solati and Soleimani, 2010a, 2010b, Hemmati *et al.*, 2015). Recent studies have shown that *jujube* has anti-cancer, anti-inflammatory, hepato-protective, anti-oxidative and anti-insomnia properties, protects the digestive system and stimulates the immune system (Guo *et al.*, 2015).

In addition, *Angelicae sinensis extract* (ASE) and *Astragalus* (AS) are commonly used in treating the age-

related diseases, which have been demonstrated in stimulating bone cell proliferation, increasing bone formation and reducing bone re-sorption in patients (Yang *et al.*, 2002). ASE induces the proliferation of ICR murine bone marrow mononuclear cells by activating ERK1/2 and P38 MAPK protein (Chen *et al.*, 2006). Feeding herbal additive was beneficial in improving average weekly body weight gain, feed conversion ratio and blood biochemical profile (Kale *et al.*, 2016). Astragalus polysaccharides (APS) have been most widely studied, for their immune potentiating properties, such as stimulation of murine cytokine production and B-cell proliferation cytokine production (Shao *et al.*, 2004). Clinical studies showed that APS could counteract the side effects of chemotherapeutic drugs and improved medulla-suppressed of cancer patient to some extent (Tin *et al.*, 2007).

Previous research showed ZJM had effects on raising the lipid metabolism and immunity function in mice/human (Ref). However, reports regarding its effects on increasing lipid metabolism and immunity function are scanty in chickens. In this study, according to the measurement of lipid metabolism and immunity function of AM, ASE and ZJM in chickens, the best way and dosage were selected and discussed the mechanisms on enriching lipid metabolism and immunity function.

MATERIALS AND METHODS

Treatment of birds

The experiment was carried out on Department of Animal Medicine, Hebei North University. A total of 400, 60-week old Hy-line laying hens were purchased from Hebei Laboratory Animal Center and housed in cages. In addition, a total of 500, 28-day old Hy-line chickens were purchased from Hebei Laboratory Animal Center and housed in cages. The hens were given free access to feed and water. All the experimental birds were treated in accordance with the guidelines of the Chinese Council for Animal Care. The experiment lasted for 42 days.

Preparation of reagent

ZJM, AS, ASE were purchased from Beijing Biochem Co., Ltd. (China). ZJM, AS and ASE were smashed to 200 mesh sieve.

Experimental design

Experiment 1 - the 60-week old Hy-line laying hens were randomly divided into ten groups with the same number and similar body weight. The hens in group1, 2 and 3 were given three gradient dosages (10 mg/kg, 20 mg/kg, 30 mg/kg) of AS respectively. Hens in group4, 5 and 6 were given three gradient dosages (10 mg/kg, 20 mg/kg, 30 mg/kg) of ASE respectively. In group 7, 8 and 9 hens were given ZJM by the same gradient dosages respectively. Hens in group 10 were fed basal diet as control. Experiment 2 - the 28-day old Hy-line chickens were also randomly divided into ten groups. The grouping scheme was the same as in experiment 1. All herbs were given for 42 days to all groups.

Blood collection and examination

At 35 and 42 days of test, one blood sample was collected from wing-vein of every replication per group to measure lipid metabolism and immunoglobulin tests. Blood samples were centrifuged for 10 min ($3,000 \times g$) at 4°C and then the pure serum samples were aspirated by pipette, stored in sterilized 1.5-mL Eppendorf tubes at -20°C and thawed at 4°C before lipid metabolism and immunoglobulin analysis. TC, TG, HDL-C and LDL-C in it and serum were quantified by colorimetry according to the protocols of commercial kits (Jiancheng Bioengineering Institute, Nanjing, China). Immunoglobulins were measured by double antibody sandwich ELISA method using commercial kits from Jiancheng Bioengineering Institute (Nanjing, China) according to the manufacturer's instructions (Geng *et al.*, 2018; Kumar *et al.*, 2018). At 42 days of test, 1 chicken was slaughtered for every replication per group to measure the immune organ indexes.

Statistical analyses

All data had a normal distribution and presented as mean $\pm$ standard deviation (SD) and analyzed by SPSS19.0 statistical software. Statistical significance was determined by one-way analysis of variance (ANOVA). A probability of less than 0.01 was considered to be statistically significant.

RESULTS AND DISCUSSION

Effects of Chinese herbal medicines on lipid metabolism in laying hens

It was shown in Table 1, ZJM (30 mg/kg) could significantly increase the count of LDL-C and HDL-C and reduce the content of TC, TG than control group10 ($p < 0.01$, $p < 0.05$). ZJM (20 mg/kg) had significant effects on the count of TC, TG and LDL-C than control group10 ($p < 0.01$, $p < 0.05$). ZJM (10 mg/kg) could reduce the count of TG ($p < 0.01$). ASE (20, 30) mg/kg could significantly reduce the count of TG ($p < 0.05$). AS (20 mg/kg) could significantly reduce the count of LDL-C than control group10 ($p < 0.05$). To sum up, the effects in ZJM (30 mg/kg) were better than other groups on lipid metabolism.

Effects of Chinese herbal medicines on immunoglobulin in laying hens

As in Table 2, ZJM (30 mg/kg) could significantly increase the count of IgG compared to control 10 ($p < 0.05$). There were no differences for IgG among all other groups than the control ($p > 0.05$). However, there was no more effect on the IgA count in all the groups than control group10 ($p > 0.05$).

Effects of Chinese herbal medicines on immune organ index in chickens

There was a significant increase on the Thymus index, Spleen index and Bursa index in group 9 (ZJM, 30 mg/kg) than the control group ($p < 0.05$). ZJM in group 8 (20 mg/kg) had only significant effects on the Thymus index and Bursa index than control group10 ($p < 0.05$). However, there was

Table 1: Effects of Chinese herbal medicines on lipid metabolism in laying hens. (mmol/L)

Group	N	Dosage (mg/kg)	TC	TG	LDL-C	HDL-C
Low dosage of AS	8	10	2.74±0.11	5.50±0.07	0.69±0.05	2.04±0.05
Middle dosage of AS	8	20	2.71±0.09	5.24±0.12	0.54±0.03*	2.06±0.07
High dosage of AS	8	30	2.69±0.08	5.31±0.14	0.70±0.04	2.07±0.05
Low dosage of ASE	8	10	2.70±0.09	5.47±0.08	0.74±0.05	2.06±0.04
Middle dosage of ASE	8	20	2.62±0.11	4.45±0.13*	0.76±0.06	2.10±0.09
High dosage ASE	8	30	2.60±0.12	4.39±0.09*	0.77±0.04	2.09±0.08
Low dosage of ZJM	8	10	2.58±0.14	3.61±0.10**	0.77±0.05	2.11±0.12
Middle dosage of ZJM	8	20	2.49±0.07*	4.32±0.12*	1.03±0.04**	2.14±0.11
High dosage of ZJM	8	30	2.45±0.09*	2.93±0.07**	0.96±0.06**	2.31±0.08*
Control	8	0	2.75±0.12	5.59±0.05	0.75±0.05	2.05±0.06

**Superscript differs significantly ($P < 0.01$) compared with control group.

*superscript differs notably ($P < 0.05$) compared with control group.

Table 2: Effects of Chinese herbal medicines on immunoglobulin in laying hens. (ng/mL)

Group	N	Dosage (mg/kg)	IgG	IgA
Low dosage of AS	8	10	3148.21±102.13	915.62±42.03
Middle dosage of AS	8	20	3153.89±76.85	921.06±38.56
High dosage of AS	8	30	3141.55±84.22	926.18±48.21
Low dosage of ASE	8	10	3130.73±104.94	918.79±42.69
Middle dosage of ASE	8	20	3146.57±89.64	914.53±37.05
High dosage ASE	8	30	3175.84±107.58	928.11 ±50.14
Low dosage of ZJM	8	10	3165.32±95.60	936.24±44.52
Middle dosage of ZJM	8	20	3208.61±177.32	948.92 ±52.06
High dosage of ZJM	8	30	3297.15±126.52*	955.14 ±43.23
Control	8	0	3125.22±109.88	910.21±34.06

*superscript differs notably ($P < 0.05$) compared with control group.

no more effect on the immune organ index in all other groups than control group10 ($p > 0.05$) (Table 3). In a word, the different dosages of single AS and ASE had no effects on the immune organ index, but ZJM (20 mg/kg) could improve the immune organ index.

The results of this study clearly demonstrated that ZJM (30 mg/kg) could not only increase lipid metabolism by increasing the count of LDL-C and HDL-C, reducing the content of TC, TG but also improve immunity function by increasing the count of IgG and the immune organ index (Table 1-3). Only AS or ASE had no more effect. At the same time, low dosage of ZJM were not suitable.

Triglycerides (TG), total cholesterol (TC), high-density lipoprotein cholesterol (HDL-C), low-density lipoprotein cholesterol (LDL-C) in serum are very important indexes reflecting the lipid metabolism level of body. TG are the best form of oxidative energy supply and energy storage in animals. TC mainly exists in the form of LDL-C and HDL-C in serum. The function of HDL-C is to transport cholesterol to the liver for metabolism. The role of LDL-C is to transport endogenous cholesterol of liver synthesis to tissues. This fruit of ZJM contains high levels of calcium, phosphorus and vitamins A, C and B. Several constituents, which include flavonoids, procyanidin B2, saponins, tannins, some phenolic compounds and polysaccharides, have been

identified in ZJM (Choi *et al.*, 2011). Antioxidants and phytochemicals can improve blood lipids. Several studies have shown the effects of this fruit on obesity and hyperlipidaemia (Sabzghabae *et al.*, 2013), liver disorders (Shen *et al.*, 2009), hypertension (Koffi *et al.*, 2008) and infectious disease. Improvement of lipid profile and body weight was observed with the consumption of ZJM powder in healthy volunteers after 3 months (Mostafa and Labban, 2013). In this experiment, the results indicated that ZJM (20 and 30 mg/Kg) could reduce the count of TC, TG and increase the content of LDL-C. Only ZJM (30 mg/Kg) could increase the content of HDL-C. ASE (20 mg/Kg) and (30 mg/Kg) could reduce the count of TG. AS (20mg/Kg) could reduce the count of LDL-C. It may be phytochemicals that can cause the inhibition of lipid peroxidation, LDL-C oxidation and the rate-limiting enzyme in cholesterol synthesis.

Humoral immunity is a kind of important specific immune response mediated by B lymphocytes and also one of the main factors in resisting infectious diseases. There are three kinds of immunoglobulins in poultry: IgG, IgA and IgM (Ref.). Angelica polysaccharide can promote cellular immunity and then promote the immune response of lymphocyte. It can be seen that Angelica polysaccharide not only promote T lymphocytes to produce IL-2 but also increase macrophage to secrete TNF- α . AP could strengthen secondary erythrocyte

Table 3: Effects of Chinese herbal medicines on immune organ index in laying hens.

Group	N	Dosage (mg/kg)	Thymus index	Spleen index	Bursa index
Low dosage of AS	10	10	3.59±0.26	1.48±0.03	2.25±0.17
Middle dosage of AS	10	20	3.72±0.24	1.62±0.04	2.26±0.14
High dosage of AS	10	30	3.62±0.25	1.56±0.06	2.31±0.15
Low dosage of ASE	10	10	3.66±0.18	1.59±0.07	2.30±0.09
Middle dosage of ASE	10	20	3.68±0.29	1.71±0.04	2.27±0.17
High dosage ASE	10	30	3.71±0.24	1.64±0.05	2.29±0.18
Low dosage of ZJM	10	10	3.70±0.22	1.65±0.05	2.35±0.14
Middle dosage of ZJM	10	20	4.11±0.27*	1.68±0.04	2.53±0.15*
High dosage of ZJM	10	30	3.94±0.25*	1.76±0.06*	2.48±0.16*
Control	10	0	3.52±0.21	1.41±0.04	2.21±0.14

*Superscript differs notably ($P < 0.05$) compared with control group.

immunity and affect the coordination of immune function directly or indirectly. In this experiment, only AM or only ASE had no more effect on the immunity. ZJM (30 mg/Kg) could increase the IgG content. ZJM (30 mg/Kg) could increase the immune organ index in chickens. ZJM (20 mg/Kg) could only increase the Thymus index and Bursa index. The mechanism may be that ZJM contains high levels of vitamins C and polysaccharides, which increase immune organ growth and then strengthen immunity at a dosage of 30 mg/kg.

CONCLUSION

The results obtained from this research suggest that ZJM (30 mg/kg) could not only increase lipid metabolism by increasing the count of LDL-C and HDL-C, reducing the content of TC, TG but also improve immunity function by increasing the count of IgG and the immune organ indexes.

ACKNOWLEDGEMENT

This study was supported by the Special Funds for Layer and Broiler Innovation Team of the Second Phase of Agricultural Modernization Industry System of Hebei Province (Specialist Position for the Prevention and Control of Broiler Disease) (No.HBCT2018150207), Shandong Key Research and Development Program, China (No. 2019GNC106074).

REFERENCES

- Chen, X.P., Chen, J.H., Zhang, P., Du, J.Z. (2006). *Angelica* stimulates proliferation of murine bone marrow mononuclear cells by the MAPK pathway. *Blood Cells Molecules and Diseases*. 36: 402-405.
- Chen, Z.Y., Jiao, R., Ma, K.Y. (2008). Cholesterol-lowering nutraceuticals and functional foods. *Journal of Agricultural and Food Chemistry*. 56: 8761-8773.
- Choi, S.H., Ahn, J.B., Kozukue, N., Levin, C.E., Friedman, M. (2011). Distribution of free amino acids, flavonoids, total phenolics and antioxidative activities of *jujube* (*Ziziphus jujuba*) fruits and seeds harvested from plants grown in Korea. *Journal of Agricultural and Food Chemistry*. 59(12): 6594-6604.
- Deogburn, R., Leite, J.O., Ratliff, J., Volek, J.S., McGrane, M.M., Fernandez, M.L. (2012). Effects of increased dietary cholesterol with carbohydrate restriction on hepatic lipid metabolism in guinea pigs. *Comparat Med*. 62: 109-115.
- Hemmati, M., Asghari, S., Zohoori, E., Karamian, M. (2015). Hypoglycemic effects of three Iranian edible plants; *jujube*, *barberry* and *saffron*: correlation with serum adiponectin level. *Pakistan Journal of Pharmaceutical Science*. 28(6): 2095-2099.
- Geng, Y., Ma, Q., Wang, Z., Guo, Y. (2018). Dietary vitamin D3 supplementation protects laying hens against lipopoly saccharide induced immunological stress. *Nutrition Metabolism and Cardiovascular Diseases (Lond)* 15: 58.
- Glombitza, K., Mahran, G., Mirhom, Y., Michel, K., Motawi, T. (1994). Hypoglycemic and antihyperglycemic effects of *Zizyphus spina-christi* in rats. *Planta Medica*. 60(3): 244-247.
- Guo, S., Duan, J. A., Qian, D., Tang, Y., Wu, D., Su, S., Zhao, Y. (2015). Content variations of triterpenic acid, nucleoside, nucleobase and sugar in *jujube* (*Ziziphus jujuba*) fruit during ripening. *Food Chemistry*. 167: 468-474.
- Kale, V.R., Bhattacharya, T.K., Wankhede, S.M., Patil, C.S., Share, A.A. (2016). Effect of supplementation of *Withania somnifera* (Ashwagandha) root powder as feed additive on performance and blood biochemicals of broilers. *Indian Journal of Animal Research*. 50(1): 53-56.
- Koffi, A., Traore, F., Adjougoua, A., Diafouka, F. (2008). Effets pharmacologiques de *Ziziphus mauritiana* Lam. (Rhamnacees) sur la pression arterielle de lapin. *Phytotherapie*. 6(4): 219-227.
- Kumar, D., Kumar, S., Gera, S. and Yadav, J. (2018). Comparative assessment of biochemical parameters of Harijana, Hardhenu and Sahiwal breeds. *Indian Journal of Animal Research*. 52: 1565-1568.
- Liu, X., Byrd, J.A., Farnell, M., Ruiz-Feria, C.A. (2014). Arginine and vitamin E improve the immune response after a *Salmonella* challenge in broiler chicks. *Poultry Science*. 93: 882-890.
- Manikandan, R., Anand, A.V., Sampathkumar, P., Manoharan, N. (2017). Protective effect of *psidium guajava* leaf ethanolic extract against streptozotocin-induced diabetes and lipodosis in rats. *Indian Journal of Animal Research*. 52(8): 1198-1205.
- Mostafa, U.E.S., Labban, L. (2013). Effect of *Zizyphus jujuba* on serum lipid profile and some anthropometric measurements. *Advancement in Medicinal Plant Research*. 1(3): 49-55.
- Naftali, T., Feingelernt, H., Lesin, Y., Rauchwarger, A., Konikoff, F.

- M. (2009). *Ziziphus jujuba* extract for the treatment of chronic idiopathic constipation: a controlled clinical trial. *Digestion*. 78(4): 224-228.
- Sabzghabae, A. M., Khayam, I., Kelishadi, R., *et al.* (2013). Effect of *Zizyphus jujuba* fruits on dyslipidemia in obese adolescents: a triple-masked randomized controlled clinical trial. *Mediev Archaeol*. 67(3): 156-159.
- Shao, B.M., Xu, W., Dai, H., Tu, P.F., Li, Z.J. (2004). A study on the immune receptors for polysaccharides from the roots of *Astragalus membranaceus*, a Chinese medicinal herb. *Biochemical and Biophysical Research Communications*. 320: 1103-1111.
- Shen, X., Tang Y., Yang, R., Yu, L., Fang, T., Duan, J.A. (2009). The protective effect of *Zizyphus jujube* fruit on carbon tetrachloride-induced hepatic injury in mice by antioxidant activities. *Journal of Ethnopharmacology*. 122 (3): 555-560.
- Solati, J., Soleimani, N. (2010a). Antidiabetic effects of ethanolic extract of *Ziziphus vulgaris* L. in streptozocin-induced. *Journal of Physiol Pharmacol*. 14(2): 174-180.
- Solati, J., Soleimani N. (2010b). Antihyperglycemic and antihyperlipidemic effects of *Ziziphus vulgaris* L. on streptozocin-induced diabetic adult male Wistar rats. *Acta Diabetologica*. 47(1): 219-223.
- Sozen, E. and N.K. Ozer. (2017). Impact of high cholesterol and endoplasmic reticulum stress on metabolic diseases: an updated mini-review. *Redox Biology*. 12: 456-461.
- Sun, J.H., Monagas, M.S., Jang, A., Molokin, J.M., Harnly, J.F., Urban, G., Solano-Aguilar, Chen, P. (2015). A high fat, high cholesterol diet leads to changes in metabolite patterns in pigs: a metabolomic study. *Food Chemistry*. 173: 171-178.
- Teck, C.L., Tan, B.K., Foo, H.L., Norhani A. and Zulkifli, I. (2011). Relationships of plasma and very low density lipoprotein lipids and subfractions with abdominal fat in chickens. *Asian-Australasian Journal of Animal Sciences*. 24: 82-87.
- Tin, M., Cho, C.H., Chan, K., James, A.E., Ko, J. (2007). *Astragalus* saponins induce growth inhibition and apoptosis in human colon cancer cells and tumor xenograft. *Carcinogenesis*. 28: 1347-1355.
- Wen, C.L., W. Yan, J.X., Zheng, C.L., Ji, D. X., Zhang, C.J., Sun, N. (2018). Feed efficiency measures and their relationships with production and meat quality traits in slower growing broilers. *Poultry Science*. 97: 2356-2364.
- Yang, Q., Populo, S.M., Zhang, J., Yang, G., Kodama, H. (2002). Effect of *Angelica sinensis* on the proliferation of human bone cells. *Clinica Chimica Acta*. 324: 89-97.